

Foundations, Units & Constants (Ch. 1)

$R_u = 8.314 \text{ kJ}/(\text{kmol} \cdot \text{K})$ – universal gas constant

$g = 9.807 \text{ m/s}^2$ – standard gravity, $1 \text{ atm} = 101.325 \text{ kPa}$

$0^\circ\text{C} = 273.15 \text{ K}$, $1 \text{ kWh} = 3600 \text{ kJ}$, $\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2\text{K}^4)$ – Stefan–Boltzmann const.

$F = ma$, $W = mg$ – weight is local gravitational force, mass is constant

$v = \frac{V}{m} = \frac{1}{\rho}$ – specific volume; intensive property, mass-independent

$P = \frac{F}{A}$, $P_{abs} = P_{atm} + P_{gage} = P_{atm} - P_{vac}$ – always use absolute P in property tables/EOS

$P = P_0 + \rho gh$ – pressure increases linearly with depth h in a static fluid

$T(\text{K}) = T(^{\circ}\text{C}) + 273.15$, $T(^{\circ}\text{R}) = 1.8 T(\text{K}) = T(^{\circ}\text{F}) + 459.67$ – always use absolute T in gas laws

Terms: system/surroundings/boundary; closed (fixed mass) vs. open/CV (mass crosses boundary); intensive vs. extensive property; state fixed by properties; quasi-equilibrium process passes through equilibrium states.

Energy, Work & Heat Transfer (Ch. 2)

$KE = \frac{1}{2}mV^2$, $PE = mgz$ – mechanical energy forms

$e_{mech} = \frac{P}{\rho} + \frac{V^2}{2} + gz$ – mechanical energy per unit mass of a flowing fluid

$W = \int F ds$ – sign convention: $W > 0$ work done BY system, $Q > 0$ heat INTO system

$W_b = \int_1^2 P dV$ – boundary work = area under path on P - V diagram

$W_b = P(V_2 - V_1)$ – boundary work, constant P

$Pv^n = \text{const}$, $W_b = \frac{P_2 V_2 - P_1 V_1}{1 - n}$ ($n \neq 1$); for $n = 1$: $W_b = P_1 V_1 \ln(V_2/V_1)$ – polytropic boundary work

$\dot{Q}_{cond} = -kA \frac{dT}{dx}$, $\dot{Q}_{conv} = hA(T_s - T_\infty)$, $\dot{Q}_{rad} = \varepsilon \sigma A(T_s^4 - T_{surr}^4)$ – conduction, convection, radiation

$\dot{W} = \frac{dW}{dt}$, $\dot{Q} = \frac{dQ}{dt}$ – rate forms; $1 \text{ kW} = 1 \text{ kJ/s}$

Properties: p - v - T & Ideal Gas (Ch. 3)

$x = \frac{m_g}{m_f + m_g}$, $y = y_f + x y_{fg}$ ($y = v, u, h, s$) – quality, valid only inside the saturation dome

$Pv = RT$, $PV = mRT$, $PV = NR_u T$, $R = \frac{R_u}{M}$ – ideal gas EOS; good at low P /high T vs. critical point

$R_{air} = 0.2870 \text{ kJ}/(\text{kg} \cdot \text{K})$, $c_p = 1.005$, $c_v = 0.718 \text{ kJ}/(\text{kg} \cdot \text{K})$, $k = 1.4$, $M = 28.97 \text{ kg}/\text{kmol}$ – air reference properties

$Z = \frac{Pv}{RT}$ – compressibility factor ($Z = 1$ ideal); use $P_R = P/P_{cr}$, $T_R = T/T_{cr}$ on generalized chart

$du = c_v dT$, $dh = c_p dT$, $c_p - c_v = R$, $k = \frac{c_p}{c_v}$ – ideal gas specific heat relations (u, h functions of T alone)

$Pv^n = \text{const}$, $\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{n-1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}}$ – polytropic: $n=0$ isobaric, $n=1$ isothermal, $n=k$ isentropic, $n \rightarrow \infty$ isochoric

$v \approx v_f(T)$, $\Delta u \approx c \Delta T$, $\Delta h \approx c \Delta T + v \Delta P$ – incompressible (liquid/solid) approximation, $c_p \approx c_v \approx c$

First Law: Closed Systems & Cycles (Ch. 4)

$E_{in} - E_{out} = \Delta E_{system}$ – general statement of the first law

$Q - W = \Delta U + \Delta KE + \Delta PE$; for a stationary system $Q - W = \Delta U$

$Q - W_b = mc_v(T_2 - T_1)$ (const. V , $W_b = 0$); $Q = mc_p(T_2 - T_1)$ (const. P) – ideal gas, const. specific heats

$\dot{Q} - \dot{W} = \frac{dE_{sys}}{dt}$ – rate form, for transient/unsteady processes

$Q = 0 \Rightarrow -W = \Delta U$ (adiabatic); $\Delta U = 0 \Rightarrow Q = W$ (isothermal, ideal gas)

$W_{net} = Q_{in} - Q_{out} = \oint \delta W = \oint P dV$ – cycle net work ($\Delta U_{cycle} = 0$)

$\eta_{th} = \frac{W_{net,out}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$ – general thermal efficiency of a power cycle

Control Volume Energy Analysis (Ch. 5)

$\frac{dm_{CV}}{dt} = \sum \dot{m}_{in} - \sum \dot{m}_{out}$ – conservation of mass, general; steady flow: $\sum \dot{m}_{in} = \sum \dot{m}_{out}$

$\dot{m} = \rho A V_{avg} = \frac{A V_{avg}}{v}$ – mass flow rate

$\theta = h + \frac{V^2}{2} + gz$ – energy per unit mass of a flowing stream (h bundles u with flow work)

$$\dot{Q}_{in} - \dot{W}_{out} = \sum \dot{m}_{out} \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right) - \sum \dot{m}_{in} \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right)$$

– steady-flow energy equation (SFEE); divide by $\dot{m}$ for single inlet/exit

$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$ – nozzle/diffuser ($\dot{Q} \approx 0$, $W = 0$, $\Delta z \approx 0$); nozzle accelerates ($P \downarrow$), diffuser decelerates ($P \uparrow$)

$\dot{W}_{T,out} = \dot{m}(h_1 - h_2)$, $\dot{W}_{C,in} = \dot{m}(h_2 - h_1)$ – turbine/compressor, $\Delta KE, \Delta PE$ negligible, adiabatic

$h_1 = h_2$ – throttling valve; large ΔP drop, no work; ideal-gas T unchanged

$\sum \dot{m}_{in} h_{in} = \sum \dot{m}_{out} h_{out}$ ($\dot{W} = 0$, $\dot{Q} = 0$) – heat exchanger/mixing chamber energy balance

Second Law of Thermodynamics (Ch. 6)

Kelvin–Planck: no heat engine converts all received heat to work – some must be rejected to a sink. *Clausius:* no device transfers heat low-to-high T with no other effect (no work input).

$\eta_{th} = \frac{W_{net,out}}{Q_H} = 1 - \frac{Q_L}{Q_H}$ – thermal efficiency of a heat engine

$COP_R = \frac{Q_L}{W_{net,in}} = \frac{1}{Q_H/Q_L - 1}$, $COP_{HP} = \frac{Q_H}{W_{net,in}} = \frac{1}{1 - Q_L/Q_H} = COP_R + 1$ – refrigerator/heat pump COP

Reversible process: system & surroundings both restorable to initial state, no trace. Irreversibilities: friction, unrestrained expansion, finite- ΔT heat transfer, mixing, non-quasi-equilibrium compression/expansion.

Carnot cycle: 2 reversible isothermal + 2 reversible adiabatic processes. *Carnot principles:* no engine beats a reversible engine between the same reservoirs; all reversible engines between the same two reservoirs share the same efficiency.

$\left(\frac{Q_H}{Q_L}\right)_{rev} = \frac{T_H}{T_L}$, $\eta_{Carnot} = 1 - \frac{T_L}{T_H}$ – max possible efficiency between reservoirs at T_H, T_L (absolute)

$$COP_{R,rev} = \frac{1}{T_H/T_L - 1}, \quad COP_{HP,rev} = \frac{1}{1 - T_L/T_H} - \text{Carnot (upper-limit) COP}$$

Entropy (Ch. 7)

$$\oint \frac{\delta Q}{T} \leq 0, \quad dS = \left(\frac{\delta Q}{T} \right)_{rev}, \quad \Delta S \geq \int_1^2 \frac{\delta Q}{T} - \text{Clausius inequality; equality only if reversible}$$

$T ds = du + P dv$, $T ds = dh - v dP - T ds$ (Gibbs) relations, valid for any simple compressible substance

$s_2 - s_1 = s_2^\circ - s_1^\circ - R \ln \frac{P_2}{P_1}$ - ideal gas entropy change, variable specific heats (tabulated $s^\circ(T)$)

$s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1} = c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$ - ideal gas, constant specific heats

$s_2 - s_1 = c \ln \frac{T_2}{T_1}$ - incompressible substance ($v \approx \text{const.}$)

$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{k-1} = \left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}}, \quad \frac{P_2}{P_1} = \left(\frac{v_1}{v_2} \right)^k$ - isentropic, ideal gas, constant k (polytropic w/ $n = k$)

$\frac{P_2}{P_1} = \frac{P_{r2}}{P_{r1}}, \quad \frac{v_2}{v_1} = \frac{v_{r2}}{v_{r1}}$ - isentropic, variable specific heats (P_r, v_r tabulated vs. T)

$\eta_T = \frac{w_a}{w_s} = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$ - isentropic efficiency, turbine & nozzle (a =actual, s =isentropic exit)

$\eta_C = \frac{w_s}{w_a} = \frac{h_{2s} - h_1}{h_{2a} - h_1}$ - isentropic efficiency, compressor/pump (≤ 1); pump w/ incompressible liquid: $w_s = v(P_2 - P_1)$

$\Delta S_{sys} = \sum \frac{Q_k}{T_k} + S_{gen}, \quad \dot{S}_{gen} \geq 0$ - entropy balance; $S_{gen} = 0$ only if internally reversible

Gas Power Systems: Otto, Diesel, Dual & Brayton (Ch. 9)

Air-standard assumptions: working fluid is air (ideal gas); combustion $\rightarrow$ external heat addition; exhaust $\rightarrow$ heat rejection restoring initial state; all processes internally reversible. *Cold air-standard:* additionally constant c_p, c_v ($k = 1.4$).

$$r = \frac{V_{max}}{V_{min}} = \frac{V_{BDC}}{V_{TDC}} - \text{compression ratio}$$

$$MEP = \frac{w_{net}}{v_{max} - v_{min}} - \text{mean effective pressure: constant } P \text{ giving same } w_{net} \text{ over same displacement}$$

same w_{net} over same displacement

Otto (SI engines) - 1-2 isentropic compression, 2-3 const- V heat add, 3-4 isentropic expansion, 4-1 const- V heat reject:

$$\eta_{th,Otto} = 1 - \frac{T_4 - T_1}{T_3 - T_2} = 1 - \frac{1}{r^{k-1}} - \text{depends only on } r, k \text{ (cold-air-std); limited by knock in practice}$$

Diesel (CI engines) - 1-2 isentropic compression, 2-3 const- P heat add, 3-4 isentropic expansion, 4-1 const- V heat reject:

$$r_c = \frac{V_3}{V_2} = \frac{v_3}{v_2} - \text{cutoff ratio } (r_c = 1 \text{ recovers Otto})$$

$$\eta_{th,Diesel} = 1 - \frac{1}{r^{k-1}} \left[\frac{r_c^k - 1}{k(r_c - 1)} \right]$$

Dual cycle - heat added const- V (2-3) then const- P (3-4), closer to real combustion:

$$r_p = \frac{P_3}{P_2}, \quad r_c = \frac{V_4}{V_3}, \quad q_{in} = c_v(T_3 - T_2) + c_p(T_4 - T_3)$$

$$\eta_{th,Dual} = 1 - \frac{1}{r^{k-1}} \left[\frac{r_p r_c^k - 1}{(r_p - 1) + k r_p (r_c - 1)} \right]$$

$r_p = 1$ recovers Diesel, $r_c = 1$ recovers Otto.

Trend (fixed r): $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$. Increasing r raises η (all three) and MEP; increasing r_c lowers η .

Brayton (gas turbine/jet) - 1-2 isentropic compression, 2-3 const- P heat add, 3-4 isentropic expansion, 4-1 const- P heat reject:

$$r_p = \frac{P_2}{P_1}, \quad \eta_{th,Brayton} = 1 - \frac{1}{r_p^{(k-1)/k}} - \text{depends only on } r_p, k;$$

limited by max turbine-inlet T_3 in practice

$$w_{C,in} = c_p(T_2 - T_1), \quad w_{T,out} = c_p(T_3 - T_4), \quad r_{bw} = \frac{w_{C,in}}{w_{T,out}} -$$

back-work ratio, often 40-80% for gas turbines

$$\varepsilon = \frac{q_{regen,act}}{q_{regen,max}} = \frac{h_5 - h_2}{h_4 - h_2} - \text{regenerator effectiveness (5 = pre-heated compressor exit, 4 = turbine exit)}$$

$$\eta_{th,regen} = 1 - \left(\frac{T_1}{T_3} \right) r_p^{(k-1)/k} \quad (\varepsilon = 1, \text{ cold-air-std}) - \text{helps only when } r_p \text{ low enough that } T_4 > T_2$$

AER309 Basic Thermodynamics — Memorization sheet. No formula sheet is permitted on these exams (fully closed-book) — this is a study aid, not something to bring in. Condensed from the full verified reference in the Formula_Ref app.